

AI Research Compendium

A Comprehensive Overview of Artificial Intelligence,
Machine Learning, and Climate Applications

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Table of Contents

1. Introduction to Artificial Intelligence
2. Machine Learning Fundamentals and Advanced Techniques
3. Deep Learning Architectures and Applications
4. Large Language Models: Architecture, Training, and Applications
5. Retrieval-Augmented Generation: Architecture and Best Practices
6. Climate Change: Science, Impacts, and Solutions

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) represents one of the most transformative technological advances in human history. At its core, AI refers to the development of computer systems that can perform tasks that typically require human intelligence. These tasks include visual perception, speech recognition, decision-making, language translation, and pattern recognition.

The field of AI has evolved dramatically since its inception in the 1950s. Alan Turing's seminal paper "Computing Machinery and Intelligence" (1950) posed the fundamental question "Can machines think?" and introduced the Turing Test as a measure of machine intelligence. The Dartmouth Conference of 1956, organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon, is widely considered the birthplace of AI as a formal academic discipline.

Early AI research focused on symbolic reasoning and logic-based approaches. The Logic Theorist (1956), developed by Allen Newell and Herbert A. Simon, was capable of proving mathematical theorems and is considered the first AI program. Throughout the 1960s and 1970s, researchers developed expert systems that encoded human knowledge as rules, enabling computers to make decisions in specialized domains like medical diagnosis and chemical analysis. Systems like MYCIN for infectious disease diagnosis and DENDRAL for chemical analysis demonstrated that AI could perform at expert levels in narrow domains.

However, the limitations of rule-based approaches became apparent during the "AI winters" of the 1970s and 1980s, when funding and interest declined due to unmet expectations. The combinatorial explosion of rules, the difficulty of knowledge acquisition, and the brittleness of systems outside their narrow domains limited practical applications. The resurgence of AI in the 21st century has been driven by three key factors: the availability of massive datasets (big data), dramatic improvements in computational power (especially GPUs and TPUs), and algorithmic breakthroughs in machine learning and deep learning.

Today, AI systems have achieved superhuman performance in specific domains. The progression has been remarkable: IBM's Deep Blue defeated world chess champion Garry Kasparov in 1997. IBM Watson won Jeopardy against human champions in 2011. DeepMind's AlphaGo defeated world champion Lee Sedol at Go in 2016. More recently, systems like AlphaFold have solved the 50-year grand challenge of protein structure prediction, and large language models have demonstrated capabilities that many researchers thought were decades away.

The economic impact of AI is projected to be enormous. According to various estimates from McKinsey, PwC, and others, AI could contribute between \$13 trillion and \$15.7 trillion to the global economy by 2030. Industries being transformed include healthcare (diagnosis, drug discovery, personalized medicine), finance (fraud detection, algorithmic trading, risk assessment), transportation (autonomous vehicles, logistics optimization), manufacturing (predictive maintenance, quality control), education (personalized learning, automated grading), and entertainment (content recommendation, game design).

The development of artificial general intelligence (AGI) - systems that can match or exceed human cognitive abilities across a wide range of tasks - remains a long-term goal and subject of intense debate. While some experts predict AGI within decades, others argue that fundamental breakthroughs in our understanding of intelligence, consciousness, and causal reasoning are still needed. The responsible development and deployment of AI systems will be one of the defining challenges of the 21st century, requiring collaboration across disciplines and international cooperation between governments, industry, academia, and civil society.

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2. Machine Learning Fundamentals and Advanced Techniques

Machine learning is a subset of AI that focuses on building systems that learn from data rather than being explicitly programmed. The fundamental principle is that by exposing algorithms to large amounts of data, they can identify patterns, make predictions, and improve their performance over time without human intervention. This paradigm shift from rule-based programming to data-driven learning has been the engine behind most recent AI advances.

Supervised Learning forms the backbone of most commercial ML applications. In this paradigm, algorithms are trained on labeled datasets where the correct outputs are known. The algorithm learns to map inputs to outputs by minimizing the difference between its predictions and the actual labels through iterative optimization. Key algorithms include: Linear Regression for continuous value prediction, Logistic Regression for binary classification, Decision Trees and Random Forests for interpretable rule-based models, Support Vector Machines (SVMs) for maximum-margin classification, Gradient Boosting Machines (XGBoost, LightGBM, CatBoost) that dominate structured data competitions, and Neural Networks that form the foundation of deep learning.

The mathematical foundations of supervised learning involve concepts from linear algebra, calculus, probability theory, and optimization. The bias-variance tradeoff is a central concept: models with high bias underfit the data (too simple), while models with high variance overfit (too complex). Regularization techniques like L1 (Lasso) and L2 (Ridge) penalties, dropout in neural networks, and early stopping help manage this tradeoff. Cross-validation provides robust performance estimation by partitioning data into training and validation sets, with k-fold cross-validation being the most common approach.

Unsupervised Learning discovers hidden patterns in unlabeled data. Clustering algorithms like K-means partition data into groups based on similarity metrics. Hierarchical clustering builds tree-like structures showing relationships at multiple scales. DBSCAN identifies clusters of arbitrary shape and detects outliers. Dimensionality reduction techniques - PCA for linear projections, t-SNE and UMAP for nonlinear manifold learning - enable visualization and compression of high-dimensional data while preserving important structures. Gaussian Mixture Models provide probabilistic cluster assignments. Autoencoders learn compressed representations through neural network bottlenecks, with applications in anomaly detection, denoising, and generative modeling.

Reinforcement Learning (RL) represents a fundamentally different learning paradigm where an agent learns by interacting with an environment. The agent observes states, takes actions, receives rewards, and learns a policy that maximizes cumulative reward. Key algorithms include Q-learning and SARSA for tabular settings, Deep Q-Networks (DQN) that combine Q-learning with deep neural networks, Policy Gradient methods that directly optimize the policy, Actor-Critic architectures that learn both value functions and policies, and Proximal Policy Optimization (PPO) that provides stable training through clipped objectives. RL has achieved superhuman performance in Atari games, Go, chess, StarCraft II, and Dota 2, and is increasingly applied to robotics, recommendation systems, and autonomous systems.

The ML workflow involves systematic stages: problem framing (defining objectives, metrics, constraints), data collection and validation (ensuring quality, representativeness, and legal compliance), exploratory data analysis (understanding distributions, correlations, anomalies), feature engineering (creating informative representations from raw data), model selection (choosing appropriate algorithms based on problem characteristics), training and hyperparameter optimization, rigorous evaluation on held-out test sets, and deployment with monitoring for concept drift and performance degradation. MLOps practices bring DevOps principles to ML systems, enabling continuous integration, delivery, and monitoring of models in production.

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3. Deep Learning Architectures and Applications

Deep learning represents a sophisticated subset of machine learning based on artificial neural networks with multiple layers. These deep neural networks learn hierarchical representations, with each layer building upon the previous one to extract increasingly abstract features. The depth allows learning complex functions that shallow networks cannot represent efficiently, making deep learning particularly powerful for perception tasks involving images, audio, and text.

Convolutional Neural Networks (CNNs) revolutionized computer vision through their biologically-inspired architecture. Convolutional layers apply learnable filters across spatial dimensions, capturing local patterns regardless of position. Pooling layers reduce spatial dimensions, providing translation invariance and computational efficiency. The progression of CNN architectures tells a story of innovation: LeNet-5 (1998) pioneered the convolution-pooling-fully-connected pattern for digit recognition. AlexNet (2012) demonstrated the power of GPU-accelerated deep learning on ImageNet. VGGNet (2014) showed that depth matters with its uniform 3x3 convolution design. GoogLeNet/Inception introduced multi-scale processing with parallel convolution paths. ResNet (2015) solved the degradation problem with skip connections, enabling networks with over 100 layers. DenseNet, EfficientNet, ConvNeXt, and vision transformers continue pushing boundaries.

Modern CNN applications extend far beyond image classification. Object detection architectures like YOLO, SSD, and Faster R-CNN enable real-time identification and localization of multiple objects. Semantic segmentation with U-Net and DeepLab assigns class labels to every pixel, crucial for medical image analysis and autonomous driving. Instance segmentation with Mask R-CNN distinguishes individual object instances. Face recognition systems achieve over 99% accuracy on benchmark datasets. Style transfer, image inpainting, and super-resolution demonstrate creative and restorative capabilities.

Recurrent Neural Networks (RNNs) process sequential data by maintaining hidden states that capture temporal dependencies. Long Short-Term Memory (LSTM) networks introduced gating mechanisms - input gate, forget gate, output gate - that control information flow, solving the vanishing gradient problem that plagued simple RNNs. Gated Recurrent Units (GRUs) simplified the architecture while maintaining effectiveness. Bidirectional RNNs process sequences in both directions, capturing context from past and future. Sequence-to-sequence models with attention mechanisms enabled breakthroughs in machine translation, speech recognition, and text summarization.

The Transformer architecture (Vaswani et al., 2017) has become the dominant paradigm across AI. Replacing recurrence with self-attention, Transformers process entire sequences in parallel, dramatically improving training efficiency on modern hardware. Multi-head attention allows the model to attend to different representation subspaces simultaneously. Positional encodings inject sequence order information since the architecture lacks inherent sequential processing. Layer normalization and residual connections enable stable training of deep architectures. The encoder-decoder structure supports sequence transduction tasks, while decoder-only (GPT) and encoder-only (BERT) variants excel at generation and understanding respectively.

Self-supervised learning has emerged as a powerful paradigm that leverages unlabeled data. Models learn useful representations by solving pretext tasks derived from the data itself, then transfer this knowledge to downstream tasks. Contrastive learning methods like SimCLR and MoCo pull together representations of augmented views of the same instance while pushing apart others. Masked language modeling (BERT) and masked image modeling (MAE) demonstrate that predicting hidden portions of data teaches rich representations. This approach has been crucial for domains where labeled data is scarce or expensive,

enabling pretraining on vast unlabeled corpora followed by efficient fine-tuning on small labeled datasets.

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Self-supervised learning has emerged as a powerful paradigm that leverages unlabeled data. Models learn useful representations by solving pretext tasks derived from the data itself, then transfer this knowledge to downstream tasks. Contrastive learning methods like SimCLR and MoCo pull together representations of augmented views of the same instance while pushing apart others. Masked language modeling (BERT) and

masked image modeling (MAE) demonstrate that predicting hidden portions of data teaches rich representations. This approach has been crucial for domains where labeled data is scarce or expensive, enabling pretraining on vast unlabeled corpora followed by efficient fine-tuning on small labeled datasets.

Deep learning represents a sophisticated subset of machine learning based on artificial neural networks with multiple layers. These deep neural networks learn hierarchical representations, with each layer building upon the previous one to extract increasingly abstract features. The depth allows learning complex functions that shallow networks cannot represent efficiently, making deep learning particularly powerful for perception tasks involving images, audio, and text.

Convolutional Neural Networks (CNNs) revolutionized computer vision through their biologically-inspired architecture. Convolutional layers apply learnable filters across spatial dimensions, capturing local patterns regardless of position. Pooling layers reduce spatial dimensions, providing translation invariance and computational efficiency. The progression of CNN architectures tells a story of innovation: LeNet-5 (1998) pioneered the convolution-pooling-fully-connected pattern for digit recognition. AlexNet (2012) demonstrated the power of GPU-accelerated deep learning on ImageNet. VGGNet (2014) showed that depth matters with its uniform 3x3 convolution design. GoogLeNet/Inception introduced multi-scale processing with parallel convolution paths. ResNet (2015) solved the degradation problem with skip connections, enabling networks with over 100 layers. DenseNet, EfficientNet, ConvNeXt, and vision transformers continue pushing boundaries.

Modern CNN applications extend far beyond image classification. Object detection architectures like YOLO, SSD, and Faster R-CNN enable real-time identification and localization of multiple objects. Semantic segmentation with U-Net and DeepLab assigns class labels to every pixel, crucial for medical image analysis and autonomous driving. Instance segmentation with Mask R-CNN distinguishes individual object instances. Face recognition systems achieve over 99% accuracy on benchmark datasets. Style transfer, image inpainting, and super-resolution demonstrate creative and restorative capabilities.

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4. Large Language Models: Architecture, Training, and Applications

Large Language Models (LLMs) represent a paradigm shift in artificial intelligence, enabling machines to understand, generate, and manipulate human language with unprecedented sophistication. These models are built on the Transformer architecture and trained on vast corpora of text using self-supervised learning objectives. The scale of these models - from millions to trillions of parameters - has revealed emergent capabilities that were not explicitly programmed by their creators.

The evolution of LLMs has been remarkably rapid. Early neural language models like word2vec (2013) learned static word embeddings capturing semantic relationships. ELMo (2018) introduced contextual word representations using bidirectional LSTMs. BERT (2018) demonstrated that bidirectional pre-training with masked language modeling and next-sentence prediction produces versatile representations that could be fine-tuned for diverse tasks. GPT (2018) showed that generative pre-training on large corpora followed by task-specific fine-tuning was highly effective. GPT-2 (2019) surprised researchers with its coherent text generation capabilities, leading to staged release and discussions about responsible AI publication. GPT-3 (2020) represented a quantum leap: 175 billion parameters trained on approximately 570GB of text data. Its few-shot learning capability - adapting to new tasks from just a few examples without gradient updates - demonstrated emergent capabilities and sparked intense interest in scaling laws.

ChatGPT's release in November 2022 brought LLMs into mainstream consciousness, reaching 100 million users within two months - the fastest adoption of any consumer application in history. Built on GPT-3.5 and fine-tuned using Reinforcement Learning from Human Feedback (RLHF), ChatGPT demonstrated conversational abilities that felt natural, helpful, and safe. The RLHF process involves: (1) supervised fine-tuning on human demonstrations of desired behavior, (2) training a reward model on human preference comparisons between model outputs, and (3) optimizing the language model policy against the reward model using Proximal Policy Optimization. Subsequent models including GPT-4 (multimodal, improved reasoning), Claude (focused on safety and honesty), Gemini (natively multimodal), Llama (open weights enabling research), Mistral (efficient architectures), and DeepSeek (strong reasoning at lower cost) have continued pushing boundaries.

The technical architecture of modern LLMs involves several key components. Tokenization converts raw text into model-digestible units: Byte-Pair Encoding (BPE) iteratively merges frequent character pairs, SentencePiece treats text as a sequence of Unicode characters, and tokenizer choice significantly impacts multilingual performance, code handling, and efficiency. Embedding layers map tokens to dense vectors that capture semantic relationships. Attention mechanisms compute contextual representations by weighting the importance of all tokens in the sequence. Multi-head attention runs multiple attention operations in parallel, each focusing on different aspects. FlashAttention algorithms reduce memory footprint from quadratic to linear in sequence length through tiling and recomputation strategies.

Training LLMs requires massive computational resources and sophisticated distributed systems. Models are trained on clusters of thousands of GPUs or TPUs for weeks to months. Distributed training strategies include data parallelism (replicating the model, splitting batches), model parallelism (splitting layers across devices), tensor parallelism (splitting individual operations), pipeline parallelism (splitting layer sequences), and ZeRO optimization (partitioning optimizer states, gradients, and parameters). Mixed precision training using bfloat16 or float16 reduces memory and increases throughput. Gradient checkpointing trades computation for memory by recomputing activations during backpropagation. 3D parallelism combines multiple strategies for optimal scaling across thousands of accelerators.

Retrieval-Augmented Generation (RAG) has emerged as a crucial pattern for grounding LLM outputs in

external knowledge. By retrieving relevant documents from a knowledge base and including them in the model's context, RAG reduces hallucinations, enables access to up-to-date information, and allows LLMs to work with proprietary or private data without fine-tuning. The approach is fundamental to building trustworthy AI applications in enterprise settings and represents a practical bridge between pure language modeling and knowledge-intensive applications that require factual accuracy and verifiability.

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5. Retrieval-Augmented Generation: Architecture and Best Practices

Retrieval-Augmented Generation (RAG) is a hybrid AI architecture that combines the strengths of information retrieval systems with the generative capabilities of large language models. First introduced by Lewis et al. at Facebook AI Research in 2020, RAG addresses fundamental limitations of standalone LLMs: knowledge cutoff dates (models only know what was in their training data), hallucinations (generating plausible but incorrect information), and the inability to access proprietary, recent, or domain-specific information without expensive fine-tuning.

The RAG indexing pipeline begins with document processing. Documents in various formats (PDF, DOCX, HTML, Markdown, plain text) must be parsed to extract clean text content. Libraries like LiteParse, Unstructured, PyPDF2, and python-docx handle different formats. The extracted text undergoes cleaning to remove artifacts, normalize whitespace, and handle special characters. The cleaned text is then split into chunks - a critical decision that significantly impacts retrieval quality. Chunking strategies include: fixed-size splitting (simple but may break sentences or thoughts mid-way), recursive character splitting (attempts to split on natural boundaries like paragraphs, sentences, then words), semantic chunking (uses embedding similarity to detect topic transitions and split at natural boundaries), and agentic chunking (uses LLMs to identify coherent segments). Common chunk sizes range from 256 to 2048 tokens, with overlap (typically 10-20% of chunk size) to maintain context across boundaries.

Embedding generation converts text chunks into dense vector representations that capture semantic meaning. The choice of embedding model involves tradeoffs: Sentence Transformers models like all-MiniLM-L6-v2 (384 dimensions, fast, local) prioritize speed and privacy; OpenAI's text-embedding-3-small/large offer high quality with API access; Cohere's embed-v3 provides multilingual support; and domain-specific models fine-tuned on scientific, legal, or medical text can outperform general models on specialized content. Embedding dimensions typically range from 384 to 3072, affecting storage requirements and search speed. The embedding model should ideally be consistent between indexing and query time to ensure vector space alignment.

Vector database selection depends on deployment requirements. LanceDB offers embedded operation with no separate server, built on the Apache Arrow/Lance columnar format, supporting vector, full-text, and hybrid search. Chroma provides a simple API with both in-memory and persistent modes. Pinecone offers a fully managed cloud service with high availability. Weaviate combines vector search with graphQL and built-in modules. Qdrant emphasizes performance with Rust implementation and quantization. Milvus targets billion-scale vector search with distributed architecture. Each database offers different tradeoffs in performance, cost, features, and operational complexity that should be evaluated against specific use case requirements.

The retrieval process transforms a user query into relevant document chunks. Query embedding uses the same model as document embedding. Similarity search typically uses cosine similarity or dot product. Hybrid search combines vector similarity with keyword-based methods like BM25, capturing both semantic meaning and exact term matching - particularly important for queries involving proper nouns, codes, or specific terminology. Advanced techniques include: query rewriting (using an LLM to expand or reformulate the query for better retrieval), multi-hop retrieval (breaking complex questions into sub-questions and chaining retrievals), recursive retrieval (using initial results to inform additional searches), and metadata filtering (restricting search to documents matching specific criteria like date ranges or categories).

Evaluation of RAG systems requires metrics beyond traditional LLM benchmarks. RAGAS (RAG Assessment) provides automated evaluation using LLM-as-judge: faithfulness (is the answer grounded in

retrieved documents?), answer relevancy (does the answer address the question?), context precision (are retrieved documents relevant?), context recall (are all relevant documents retrieved?), and answer correctness (factual accuracy). Human evaluation remains important for assessing nuance, particularly in high-stakes domains like healthcare and law. A/B testing and user feedback collection enable continuous improvement in production systems, with logged queries and responses providing training data for system refinement.

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6. Climate Change: Science, Impacts, and Solutions

Climate change represents one of the most urgent challenges facing humanity in the 21st century. The scientific consensus is unequivocal: Earth's climate is warming at an unprecedented rate due to human activities, primarily the emission of greenhouse gases from burning fossil fuels, deforestation, and industrial processes. The Intergovernmental Panel on Climate Change (IPCC), representing thousands of scientists worldwide, has documented this consensus through comprehensive assessment reports spanning decades of research.

The fundamental physics of the greenhouse effect has been understood for over a century. In 1896, Swedish scientist Svante Arrhenius first calculated that doubling atmospheric CO₂ concentrations would raise global temperatures by several degrees - a remarkably accurate prediction given the tools available. Greenhouse gases (GHGs) including carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases absorb and re-emit infrared radiation, trapping heat in the atmosphere. While this effect is natural and essential for maintaining Earth's habitable temperature, human activities have dramatically amplified it. Since the Industrial Revolution (circa 1750), atmospheric CO₂ concentrations have increased from approximately 280 parts per million (ppm) to over 420 ppm - the highest level in at least 800,000 years, as determined from Antarctic ice core records that preserve ancient air bubbles.

The observed impacts of climate change are extensive and accelerating. Global average surface temperatures have risen by approximately 1.1-1.2 degrees Celsius above pre-industrial levels, with land areas warming faster than oceans and the Arctic warming at roughly twice the global average rate. This warming has triggered a cascade of environmental changes: Arctic sea ice extent has declined by approximately 40% since satellite records began in 1979; global mean sea level has risen by approximately 20-23 cm since 1880, with the rate accelerating from 1.4 mm/year over most of the 20th century to 3.6 mm/year in recent decades; glaciers worldwide are retreating, affecting water supplies for billions of people; ocean acidification has increased by about 30% since the Industrial Revolution, threatening marine ecosystems and coral reefs; and the frequency and intensity of extreme weather events - heatwaves, heavy precipitation, droughts, and tropical cyclones - have increased in many regions according to attribution studies.

Climate models project that without substantial emissions reductions, global temperatures could rise by 2.5-4.5 degrees Celsius by 2100, with catastrophic consequences for ecosystems, food security, water resources, human health, and economic stability. The 2015 Paris Agreement, signed by 196 parties, established the goal of limiting warming to well below 2 degrees Celsius, preferably 1.5 degrees, above pre-industrial levels. However, current national commitments (Nationally Determined Contributions) put the world on track for approximately 2.5-2.9 degrees of warming. Bridging this gap requires rapid, far-reaching transformations across all sectors of the global economy, including energy, transportation, industry, agriculture, and land use.

Mitigation strategies focus on reducing greenhouse gas emissions and enhancing carbon sinks. The energy sector, responsible for approximately 73% of global GHG emissions, must transition from fossil fuels to renewable sources. Solar photovoltaic costs have fallen by over 90% since 2010, and wind energy costs by over 70%, making them increasingly competitive with conventional generation. Battery storage costs have declined similarly, addressing intermittency challenges. Electrification of transportation, heating, and industrial processes, coupled with clean electricity generation, is a core decarbonization pathway. Energy efficiency improvements across buildings, industry, and transportation can reduce energy demand while maintaining or improving services. Carbon dioxide removal (CDR) technologies, including afforestation/reforestation, bioenergy with carbon capture and storage (BECCS), direct air capture, and enhanced weathering, may be

necessary to achieve net-zero emissions and eventually net-negative emissions to stabilize the climate system.

Adaptation strategies address the impacts of climate change that are already unavoidable due to past and committed future emissions. Coastal communities require sea walls, mangrove restoration, and managed retreat strategies to cope with sea level rise and storm surge. Agricultural systems need drought-resistant crop varieties, improved irrigation efficiency, and diversified farming practices to maintain food security. Water management must address both increased flood risks and water scarcity through improved infrastructure, groundwater management, and water conservation. Public health systems need to prepare for increased heat-related illness, expanded ranges of vector-borne diseases, and mental health impacts of climate disasters. Early warning systems for extreme weather events save lives and reduce economic losses. Ecosystem-based adaptation leverages natural systems - wetlands for flood protection, forests for temperature regulation - providing co-benefits for biodiversity and human well-being.

International climate finance is crucial for enabling mitigation and adaptation, particularly in developing countries that have contributed least to historical emissions but face the most severe consequences. Developed countries committed to mobilizing \$100 billion annually by 2020 (met in 2022) and are negotiating a new collective quantified goal for the post-2025 period. The Loss and Damage fund, established at COP27 (2022) and operationalized at COP28 (2023), addresses impacts beyond adaptation limits. Carbon markets, including emissions trading systems and carbon offsets, provide economic mechanisms for emissions reductions, though their effectiveness depends on robust monitoring, reporting, and verification frameworks.

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